

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						***** VALVES *****			817 B1 B2 VB2	USE ONLY FOR TIE IN ALL BOLTS ARE METRIC DIA. 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
20	U04						***** MODULAR VALVES *****				
30											
30	U04		MODULAR DBB	M06AC02	0.75	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
40											
40	U04	VB2					***** BALL VALVES *****			VB2	
50											
50	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
60	U04		BALL	B06AC05	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0		
70											
70	U04						***** GATE VALVES *****				
80											
80	U04		GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
90	U04		GATE	W06AC71	6	24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
100											
100	U04						*****PIPE ABOVE GROUND*****				
110											
110	U04		PIPE	5358514	0.75	3	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
120											
120	U04		PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
130	U04		PIPE	9001255	6	6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE	0	0		
140	U04		PIPE	9001311	8	8	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
150	U04		PIPE	9001487	10	10	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		
160	U04		PIPE	9001494	12	14	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE	0	0		
170	U04		PIPE	9001490	16	16	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE	0	0		
180	U04		PIPE	9001495	18	18	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE	0	0		
190	U04		PIPE	9005244	20	20	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0		
200	U04		PIPE	9005243	24	24	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE	0	0		
210	U04		PIPE	9001306	28	28	LINEPIPE DSAW WT= 19.05MM API-5L X60 BE	0	0		
220											
220	U04						***** PIPE BELOW GROUND *****				
230											
230	U04		PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
240	U04		PIPE	9001255CTD	6	6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE COATED	0	0		
250	U04		PIPE	9001311CTD	8	8	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0		
260	U04		PIPE	9001487CTD	10	10	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
270	U04		PIPE	9001494CTD	12	14	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED	0	0		
280	U04		PIPE	9001490CTD	16	16	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED	0	0		
290	U04		PIPE	9001495CTD	18	18	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED	0	0		
300	U04		PIPE	9005244CTD	20	20	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE COATED	0	0		
310	U04		PIPE	9005243CTD	24	24	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE COATED	0	0		
320	U04		PIPE	9001306CTD	28	28	LINEPIPE DSAW WT= 19.05MM API-5L X60 BE COATED	0	0		
330											
330	U04						***** FITTINGS ABOVE & BELOW GROUND *****				
340											
340	U04		ELBOW	5392600	0.75	3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
350	U04		ELBOW	5392606	0.75	3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
360											
360	U04		TEE	5392609	0.75	3	TEE BW SCH XS LTCS A420 WPL6	0	0		
370	U04		TEE RED	5392661	0.75	2	TEE RED BW SCH XS LTCS A420 WPL6	1	3		
380	U04		TEE RED	9006028	6	6	TEE RED BW WT 11.91 MM/7.11 MM LTCS A860 WPHY 60	16	16		
390	U04		TEE RED	9006029	8	8	TEE RED BW WT 15.88 MM/7.92 MM LTCS A860 WPHY 60-W	20	20		
400											
400	U04		CAP	5392610	0.75	3	CAP BW SCH XS LTCS A420 WPL6	0	0		
410	U04		CAP	9005062	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
420	U04		CAP	9005056	6	6	CAP BW WT 7.11MM LTCS A860 WPHY-60	0	0		
430	U04		CAP	9005058	8	8	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0		
440	U04		CAP	9005059	10	10	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
450	U04		CAP	9005061	12	14	CAP BW WT 10.31 MM LTCS A860 WPHY-60	0	0		
460	U04		CAP	9005065	16	16	CAP BW WT 11.91 MM LTCS A860 WPHY-60	0	0		
470	U04		CAP	9005067	18	18	CAP BW WT 12.70 MM LTCS A860 WPHY-60	0	0		
480	U04		CAP	9005069	20	20	CAP BW WT 15.88 MM LTCS A860 WPHY-60	0	0		
490	U04		CAP	9005072	24	28	CAP BW WT 19.05 MM LTCS A860 WPHY-60	0	0		
500											
500	U04		RED CONC	5392458	0.75	2	REDUCER CONC SCH XS LTCS A420 WPL6	1	3		
510	U04		RED ECC	5392461	0.75	2	REDUCER ECC SCH XS LTCS A420 WPL6	1	3		
520	U04		RED ECC	9001372	1.5	3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
530	U04		RED ECC	9001582	3	3	REDUCER ECC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6	6		
540	U04		RED ECC	9001373	4	4	REDUCER ECC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	6	6		
550	U04		RED ECC	9001374	4	4	REDUCER ECC WT 7.92 MM / 6.02 MM LTCS A860 WPHY-60	8	8		
560	U04		RED ECC	9001375	6	6	REDUCER ECC WT 7.92 MM / 7.11 MM LTCS A860 WPHY-60	8	8		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 24" AND ABOVE SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 /100 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	600K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
570	U04		RED ECC	9001376	4	4	REDUCER ECC WT 8.74 MM / 6.02 MM LTCS A860 WPHY-60	10	10	
580	U04		RED ECC	9001377	6	6	REDUCER ECC WT 8.74 MM / 7.11 MM LTCS A860 WPHY-60	10	10	
590	U04		RED ECC	9002250	8	8	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	10	10	
600	U04		RED ECC	9001378	6	6	REDUCER ECC WT 10.31 MM / 7.11 MM LTCS A860 WPHY-60	12	14	
610	U04		RED ECC	9001379	8	8	REDUCER ECC WT 10.31 MM / 7.92 MM LTCS A860 WPHY-60	12	14	
620	U04		RED ECC	9002399	10	10	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60	12	14	
630	U04		RED ECC	9005035	12	12	REDUCER ECC WT 10.31 MM LTCS A860 WPHY-60	14	14	
640	U04		RED ECC	9005036	8	8	REDUCER ECC WT 11.91 MM / 7.92 MM LTCS A860 WPHY-60	16	16	
650	U04		RED ECC	9001380	10	10	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60	16	16	
660	U04		RED ECC	9001382	12	14	REDUCER ECC WT 11.91 MM / 10.31 LTCS A860 WPHY-60	16	16	
670	U04		RED ECC	9005116	10	10	REDUCER ECC WT 12.70 MM / 8.74 MM LTCS A860 WPHY-60	18	18	
680	U04		RED ECC	9002406	12	14	REDUCER ECC WT 12.70 MM / 10.31 MM LTCS A860 WPHY-60	18	18	
690	U04		RED ECC	9001590	16	16	REDUCER ECC WT 12.70 MM / 11.91 MM LTCS A860 WPHY-60	18	18	
700	U04		RED ECC	9006250	12	14	REDUCER ECC WT 15.88 MM / 10.31 MM LTCS A860 WPHY-60-W	20	20	
710	U04		RED ECC	9004151	16	16	REDUCER ECC WT 15.88 MM / 11.91 MM LTCS A860 WPHY-60-W	20	20	
720	U04		RED ECC	9001384	18	18	REDUCER ECC WT 15.88 MM / 12.70MM LTCS A860 WPHY-60-W	20	20	
730	U04		RED ECC	9004301	18	18	REDUCER ECC WT 19.05MM / 12.70MM LTCS A860 WPHY-60-W	24	28	
740	U04		RED ECC	9004302	20	20	REDUCER ECC WT 19.05MM / 15.88MM LTCS A860 WPHY-60-W	24	28	
750	U04		RED ECC	9006251	24	24	REDUCER ECC WT 19.05 MM LTCS A860 WPHY-60-W	28	28	
760										
760	U04						***** INDUCTION BENDS ABOVE GROUND *****			
770										
770	U04		INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
780	U04		INDUC BEND	9001396	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
790	U04		INDUC BEND	9001815	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
800	U04		INDUC BEND	9001816	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
810	U04		INDUC BEND	9001818	12	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
820	U04		INDUC BEND	9001397	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
830	U04		INDUC BEND	9005321	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
840	U04		INDUC BEND	9001398	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
850	U04		INDUC BEND	9002060	24	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
860	U04		INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
870	U04		INDUC BEND	9001399	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
880	U04		INDUC BEND	9001825	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
890	U04		INDUC BEND	9001826	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
900	U04		INDUC BEND	9001828	12	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
910	U04		INDUC BEND	9001500	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
920	U04		INDUC BEND	9005327	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
930	U04		INDUC BEND	9001200	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
940	U04		INDUC BEND	9002061	24	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
950	U04		INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
960	U04		INDUC BEND	9002000	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
970	U04		INDUC BEND	9001835	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
980	U04		INDUC BEND	9001836	10	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
990	U04		INDUC BEND	9005041	12	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1000	U04		INDUC BEND	9002001	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1010	U04		INDUC BEND	9005333	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1020	U04		INDUC BEND	9002002	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1030	U04		INDUC BEND	9002062	24	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1040										
1040	U04						***** INDUCTION BENDS BELOW GROUND *****			
1050										
1050	U04		INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1060	U04		INDUC BEND	9001396CTD	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1070	U04		INDUC BEND	9001815CTD	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1080	U04		INDUC BEND	9001816CTD	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1090	U04		INDUC BEND	9001818CTD	12	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1100	U04		INDUC BEND	9001397CTD	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1110	U04		INDUC BEND	9005321CTD	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1120	U04		INDUC BEND	9001398CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1130	U04		INDUC BEND	9002060CTD	24	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1140	U04		INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1150	U04		INDUC BEND	9001399CTD	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1160	U04		INDUC BEND	9001825CTD	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1170	U04		INDUC BEND	9001826CTD	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1180	U04		INDUC BEND	9001828CTD	12	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1190	U04		INDUC BEND	9004001CTD	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGINZCHEVROI TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 24" AND ABOVE SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 /100 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	600K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1200	U04		INDUC BEND	9005327CTD	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1210	U04		INDUC BEND	9001200CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1220	U04		INDUC BEND	9002061CTD	24	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1230	U04		INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1240	U04		INDUC BEND	9002000CTD	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1250	U04		INDUC BEND	9001835CTD	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1260	U04		INDUC BEND	9001836CTD	10	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1270	U04		INDUC BEND	9005041CTD	12	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1280	U04		INDUC BEND	9002001CTD	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1290	U04		INDUC BEND	9004000CTD	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1300	U04		INDUC BEND	9002002CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1310	U04		INDUC BEND	9002062CTD	24	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1320										
1320	U04						***** OLETS ABOVE & BELOW GROUND *****			
1330										
1330	U04		OLETS	9005117	0.75	2	WELDOLET WT 7.11MM/ SCH XS LTCS A694 F60	6	6	
1340	U04		OLETS	9002552	0.75	3	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	8	8	
1350	U04		OLETS	9006423	0.75	3	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	10	10	
1360	U04		OLETS	9002551	0.75	3	WELDOLET WT 10.31MM/ XS WT LTCS A694 F60	12	14	
1370	U04		OLETS	9006433	4	4	WELDOLET WT 10.31 MM/6.02 MM LTCS A694 F60	12	14	
1380	U04		OLETS	9002003	0.75	3	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	16	16	
1390	U04		OLETS	9002004	4	4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	16	16	
1400	U04		OLETS	9002467	0.75	3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	18	18	
1410	U04		OLETS	9005042	4	4	WELDOLET WT 12.70 MM/ 6.02 MM WT LTCS A694 F60	18	18	
1420	U04		OLETS	9005043	6	6	WELDOLET WT 12.70 MM/ 7.11 MM WT LTCS A694 F60	18	18	
1430	U04		OLETS	9005045	0.75	3	WELDOLET WT 15.88 MM/ SCH XS LTCS A694 F60	20	20	
1440	U04		OLETS	9005046	4	4	WELDOLET WT 15.88 MM/ 6.02 MM LTCS A694 F60	20	20	
1450	U04		OLETS	9002005	6	6	WELDOLET WT 15.88 MM/ 7.11 MM LTCS A694 F60	20	20	
1460	U04		OLETS	9005047	8	8	WELDOLET WT 15.88 MM/ 7.92 MM LTCS A694 F60	20	20	
1470	U04		OLETS	9006252	0.75	3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	24	28	
1480	U04		OLETS	9005386	4	4	WELDOLET WT 19.05 MM/ 6.02MM LTCS A694 F60	24	28	
1490	U04		OLETS	9002093	6	6	WELDOLET WT 19.05 MM/ 7.11 MM LTCS A694 F60	24	28	
1500	U04		OLETS	9002094	8	8	WELDOLET WT 19.05 MM/ 7.92 MM LTCS A694 F60	24	28	
1510	U04		OLETS	9002090	10	10	WELDOLET WT 19.05 MM/ 8.74 MM LTCS A694 F60	28	28	
1520										
1520	U04		OLETS	5504006	0.75	2	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2 CL.1	3	3	
1530	U04		OLETS	9005030	0.75	2	NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60	4	4	
1540	U04		OLETS	9005100	0.75	2	NIPOFLANGE 600# RF 7.11 MM/XS WT LTCS A694 F60	6	6	
1550	U04		OLETS	9005031	0.75	2	NIPOFLANGE 600# RF 7.92 MM/XS WT LTCS A694 F60	8	8	
1560	U04		OLETS	9005032	0.75	2	NIPOFLANGE 600# RF 8.74 MM/XS WT LTCS A694 F60	10	10	
1570	U04		OLETS	9005033	0.75	2	NIPOFLANGE 600# RF 10.31 MM/XS WT LTCS A694 F60	12	14	
1580	U04		OLETS	9005103	0.75	2	NIPOFLANGE 600# RF 11.91 MM/XS WT LTCS A694 F60	16	16	
1590	U04		OLETS	9005104	0.75	2	NIPOFLANGE 600# RF 12.70 MM/XS WT LTCS A694 F60	18	18	
1600	U04		OLETS	9005034	0.75	2	NIPOFLANGE 600# RF 15.88 MM/XS WT LTCS A694 F60	20	20	
1610	U04		OLETS	9003000	0.75	2	NIPOFLANGE 600# RF WT 19.05MM/SCH XS LTCS A694 F60	24	28	
1620										
1620	U04						***** FLANGES ABOVE & BELOW GROUND*****			
1630										
1630	U04		FLANGE	5557036	0.75	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1640	U04		FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
1650	U04		FLANGE	9001462	6	6	FLG WN 600# RF LTCS A694 F60 WT 7.11MM	0	0	
1660	U04		FLANGE	9003494	8	8	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM	0	0	
1670	U04		FLANGE	9003484	10	10	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0	
1680	U04		FLANGE	9003486	12	14	FLG WN 600# RF LTCS A694 F60 WT 10.31 MM	0	0	
1690	U04		FLANGE	9001511	16	16	FLG WN 600# RF LTCS A694 F60 WT 11.91MM	0	0	
1700	U04		FLANGE	9001512	18	18	FLG WN 600# RF LTCS A694 F60 WT 12.70MM	0	0	
1710	U04		FLANGE	9005128	20	20	FLG WN 600# RF LTCS A694 F60 WT 15.88 MM	0	0	
1720	U04		FLANGE	9002064	24	24	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM	0	0	
1730	U04		FLANGE	9003498	28	28	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM SERIES A	0	0	
1740	U04	817	FLANGE	9006169	28	28	FLG COMPACT 600# RJ A694 F60 WT 19.05 MM LINE CLASS 600K5E	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 24" AND ABOVE SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 /100 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	600K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1750	U04		BLIND	5615598	0.75	3	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
1760	U04		BLIND	9001518	4	24	FLG BLIND 600# RF LTCS A694 F60	0	0	
1770	U04		BLIND	9001519	28	28	FLG BLIND 600# RF LTCS A694 F60 SERIES A	0	0	
1780										
1780	U04	B1					***** GASKETS*****			
1790										
1790	U04		GASKET	5673103	0.75	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1800	U04		GASKET	5673055	28	28	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR B16.47 SERIES A	0	0	
1810	U04	817	GASKET	5673417	28	28	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 600K5F	0	0	
1820										
1820	U04						*****BOLTING*****			
1830										
1830	U04		BOLTING	5675770	0.5	28	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1840										
1840	U04		BOLTING	5675675	0.75	28	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1850										
1850	U04						***** BLINDS & SPACERS *****			
1860										
1860	U04		SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
1870										
1870	U04		PADDLE	5750040	10	28	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
1880	U04		PADDLE	5750251	10	28	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
1890										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 24" AND ABOVE SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 /100 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	600K5F